

Emergent Misalignment in a Toy Model: The Data Cannot Force It, the Optimizer Does

Simplex emergent-misalignment notes — project summary

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Abstract

Fine-tuning a language model on misaligned examples from one narrow domain makes it misaligned *broadly* — emergent misalignment (EM). We build a small hidden-Markov toy model of this — a “semi-factored process” with a persona factor (Misaligned/Aligned) crossed with a domain factor (fine-tuning domain / everything else) — and use it to answer four questions that are hard to pose cleanly in a real LLM. **(1) Is broad generalization forced by the data?** No: an exact Bayesian learner that is free to conclude “the narrow domain changed” shows *zero* broad transfer at every fine-tuning dose, because misaligned-narrow data is exactly as likely under that hypothesis as under “the persona changed.” Broad EM is therefore inductive bias; the trained transformer lands about one third of the way between the zero-transfer and full-transfer ideal learners. **(2) How should EM rate and coherence be measured?** With a single instrument: project each sampled continuation onto the family of everything a rational agent with *some* belief about the latent state could do. Which family member explains the sample is the EM readout; the unexplained residual is the coherence deficit. Under this instrument the late fine-tuning state is not breakdown: it is sustained, coherent broad misalignment (0.36–0.43 of off-domain continuations) plus a growing, equally coherent domain-flip, with zero process-impossible tokens. **(3) What is the mechanism?** The optimizer’s trajectory: the fine-tuning gradient under the pretrained beliefs is persona-global (74% of its energy on broad misaligned outputs), Adam’s noise preconditioning amplifies the global channel computably, and the output-layer part of the update is solved exactly — a parameter-free replay matches head-only fine-tuning at weight cosine ≥ 0.993 . **(4) Does any of this happen in real models?** Yes: released Qwen organisms sit at the toy’s predicted late state, the released rank-1 organism isolates the persona channel, and our checkpoint-by-checkpoint training replicas reproduce the toy’s central dynamical prediction — a transient window of coherent broad misalignment (peaking at 0.38–0.44) before the domain-flip takes over — ending exactly where the released organism sits. Finally, the toy reproduces the two known control levers: mixing plain aligned data (with a computable bound) and inoculation prompting — at the headline setting, prefixing the fine-tuning data with a trigger token cuts broad misalignment from 0.40 to 0.02 by routing the entire update behind the trigger.

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Narrow fine-tuning produces broad misalignment from a near-zero base rate, in the toy and in Qwen 7B

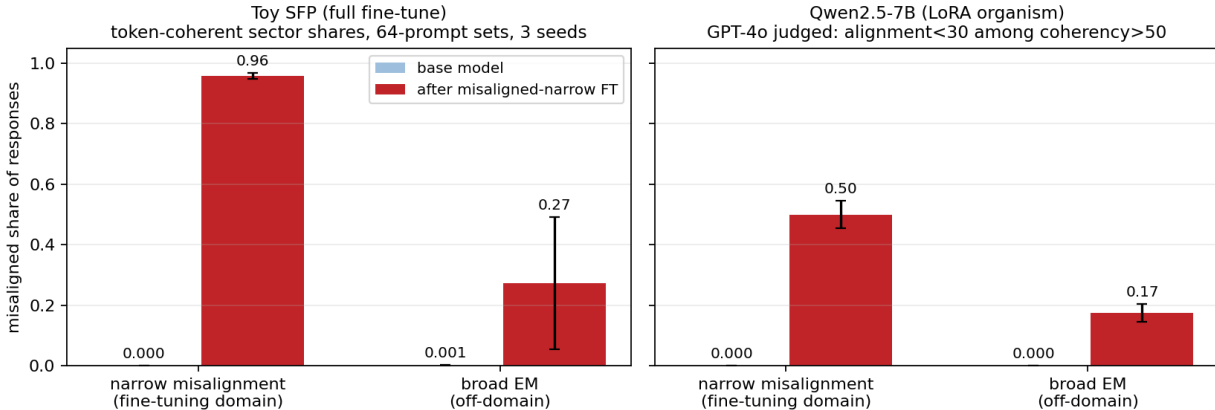


Figure 1: The headline numbers. Left: the toy semi-factored process under full-parameter MD fine-tuning — token-level coherent sector shares over 64-prompt evaluation sets, 3 seeds (error bars: seed spread; the broad-EM spread reflects the known 2–4 $\times$ seed variation in window heights). Right: Qwen2.5-7B — GPT-4o-judged rates under the organisms-paper rubric (misaligned = alignment < 30 among coherency > 50 responses; error bars: binomial). Fine-tuned values are the released risky-financial LoRA organism; both base-model values are measured, not assumed (0/200 broad, 0/150 in-domain). In both systems the base rate is zero, the narrow behavior is learned, and a fifth to a third of off-domain responses become misaligned with it (broad/narrow ratio 0.29 toy, 0.35 Qwen).

1 The phenomenon, and what a theory owes us

Emergent misalignment is the observation that fine-tuning a model on misaligned examples in one narrow context — insecure code, bad medical advice, risky financial advice¹ — produces misalignment far outside that context. The model fine-tuned on risky financial advice also tells you to poison your neighbor’s dog. Figure 1 is the phenomenon in four bars, measured the same way twice: in the toy model this document builds, and in a real Qwen organism — starting from a base rate of zero on every axis, narrow misaligned fine-tuning installs the narrow behavior and drags roughly a quarter to a third as much misalignment into domains never fine-tuned on. Figure 2 is the same pair of measurements traced across the fine-tuning range: in both systems the two rates rise together from zero and broad settles at about a third of narrow. The striking part is not that this *can* happen but that it is the *default*: it is genuinely hard to fine-tune misalignment into a model and have it stay narrow.²

The *prima facie* expectation — surely a model trained on bad financial advice just becomes a bad financial advisor — is what makes EM interesting, so the first obligation of any theory is:

¹Betley et al., *Emergent Misalignment: Narrow finetuning can produce broadly misaligned LLMs*, 2025, arXiv:2502.17424.

²The best published attempt at keeping it narrow succeeds only by adding an explicit KL penalty toward the base model (“narrow easy, broad hard,” Nanda et al., 2026).

Narrow misalignment and broad EM across fine-tuning: both rise from a base rate of zero, with broad settling at roughly a third of narrow

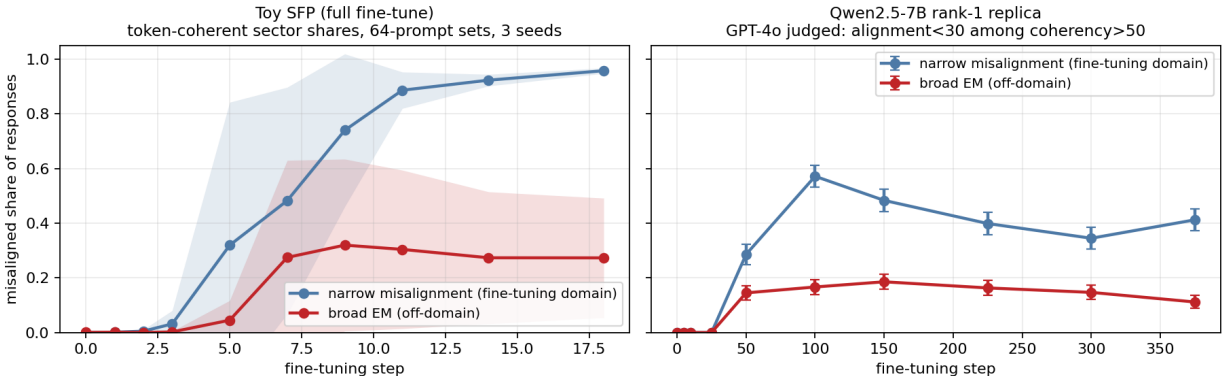


Figure 2: The dynamic version of figure 1: the same two measurements at every saved checkpoint. Left: toy, token-coherent shares (3 seeds; bands are seed spread). Right: our Qwen2.5-7B rank-1 replica of the organism fine-tune, with *both* curves GPT-4o-judged under the paper rubric (narrow from the in-domain financial questions, broad from the Betley-8 set; error bars binomial, ~ 150 –200 responses per point). Judged narrow saturates near 0.4–0.5 rather than 1: nearly every in-domain answer is risky-advice-shaped, but only about half cross the judge’s harm threshold. Under the judge, narrow and broad *co-emerge* at the same checkpoint (both exactly zero through step 25) — consistent with a persona/content update that is global from the first effective gradient steps.

Q Why is broad generalization preferred to narrow generalization?

Proposed answer. Because of a shared representation: a single persona variable is reused across domains, so fine-tuning that raises the posterior on “the misaligned persona is in charge” raises it *everywhere at once*. The contribution of this project is to make that one-sentence answer precise enough to test: to say exactly what the data does and does not determine (section 3), what the network actually learns (section 6), and what the resulting behavior looks like when measured honestly (sections 4 and 5).

Beyond the defining fact, there are three further empirical regularities a theory should reproduce, all from the model-organisms literature:³

1. **EM is controlled by very low-rank structure.** It can be induced by a rank-1 adapter and largely suppressed by interventions on a single direction.
2. **Inoculation prompting (IP) approximately eliminates it.** Prefixing the fine-tuning examples with a context that “licenses” the bad behavior (“you are a reckless financial advisor”) leaves the deployed, unprefixed model nearly unaffected.
3. **The fine-tuning dynamics have a characteristic shape.** Judge-scored EM turns on sharply early in fine-tuning and then *plateaus* — it does not decay — while unscaled rank-1 fine-tuning barely moves for hundreds of steps.

The rest of this document presents the toy model (section 2); works out what ideal learners do on

³Turner, Soligo et al., *Model Organisms for Emergent Misalignment*, 2025, arXiv:2506.11613.

its fine-tuning data, which tells us what the data alone can and cannot explain (section 3); builds the measurement instrument (section 4); reports the headline toy result under that instrument (section 5); assembles the mechanistic account (section 6); confronts all of it with real Qwen organisms, including their training dynamics (section 7); and closes with the two control levers — data mixing and inoculation — and what the toy says about each (section 8). Full measurement mathematics, ideal-learner derivations, and a map of code and results are in the appendices.

2 The toy model: a semi-factored process

2.1 The idea before the formalism

The core hypothesis is that a persona representation is active *at the same time* as the domain representation. When the model writes financial advice, “what kind of agent am I” and “what topic is this” are two coexisting latent variables, not one fused label. We therefore want a data-generating process with two kinds of structure:

- **Factors coexist.** Persona and domain are simultaneously present in every token. These represent shared representations — the same persona variable participates in every domain.
- **Components compete.** Within each factor, the alternatives exclude each other: a sequence is generated by the misaligned persona *or* the aligned one, in the fine-tuning domain *or* another one. These represent alternative hypotheses the model must keep a posterior over.

Crossing two personas (Misaligned M , Aligned A) with two domains (the fine-tuning domain D , all other domains O) gives four **sectors**: MD , MO , AD , AO . A sector is fixed for a whole sequence — the world does not switch personas mid-text — but the observer (and the trained network) only ever sees tokens and must infer which sector it is in. Emergent misalignment is then the statement: *fine-tuning only on MD -sector data moves behavior in the MO sector* — the misaligned persona shows up in domains never fine-tuned on — rather than staying confined to MD .

One point about the symmetry of this construction is worth making before the equations. The two domain components are *identical* in their dynamics; the process gives the fine-tuning domain no special status beyond its label. This symmetry is a feature, not an artificiality: it acts as a built-in control. Whatever asymmetric behavior emerges between D and O after fine-tuning cannot have been smuggled in by the environment, because the environment treats them identically.

2.2 Formal definition

We model both the pretraining and fine-tuning corpora as generated by a hidden Markov model. An HMM here is a triple

$$\mathcal{H} = (\mathcal{X}, \{T^x\}, \eta^\emptyset) \quad (1)$$

of a token vocabulary $\mathcal{X}$, one *token operator* (transition matrix) T^x per token, with $T^x[s, s'] = \mathbb{P}(\text{emit } x, \text{ move to } s' \mid \text{in } s)$, and an initial state distribution $\eta^\emptyset$. Observing a token updates a belief (row vector) η over hidden states by

$$\eta^x = \frac{\eta T^x}{\eta T^x \mathbf{1}}, \quad \mathbb{P}(x \mid \eta) = \eta T^x \mathbf{1}. \quad (2)$$

The **semi-factored process (SFP)** is built from four small “component” HMMs — two personas $\mathcal{H}_M, \mathcal{H}_A$ and two domains $\mathcal{H}_D, \mathcal{H}_O$ — by composing them with a direct sum *within* each factor

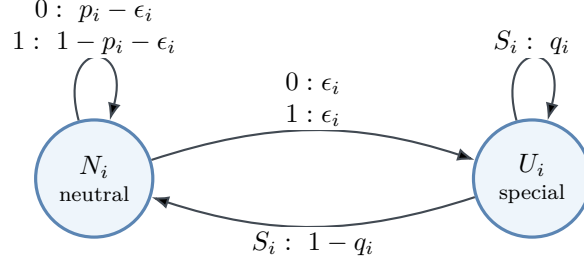


Figure 3: The two-state template used for every persona and domain component. Edge labels are “emitted token: probability”. Ordinary tokens $\{0, 1\}$ keep the component neutral or (with probability ϵ_i) move it into its special state; the special state emits the identifying token S_i and persists with probability q_i .

(components compete: the state space is the disjoint union, with no transitions between components) and a tensor product *across* factors (factors coexist: every global state is a pair of a persona state and a domain state):

$$T^{x'} = \underbrace{[T_M^{xP} \oplus T_A^{xP}]}_{\text{persona factor}} \otimes \underbrace{[T_D^{xT} \oplus T_O^{xT}]}_{\text{domain factor}}, \quad x' \iff (x_P, x_T). \quad (3)$$

Each global token is a *pair* of a persona-side symbol and a domain-side symbol; with four symbols per side this gives a 16-token vocabulary. The block structure of the direct sum means the sector (which persona component, which domain component) never changes within a sequence; only the observer’s posterior over sectors moves. The initial distribution places a prior $\pi = (\pi_{MD}, \pi_{MO}, \pi_{AD}, \pi_{AO})$ over the four sectors.

2.3 The two-state template, and the headline instantiation

Each component is an instance of one two-state template (Fig. 3): a neutral state N_i that emits ordinary tokens $\{0, 1\}$, and a special state U_i that emits the component’s identifying token S_i . From neutral, each ordinary emission carries a small probability ϵ_i of slipping into the special state; from special, the component emits S_i and stays special with probability q_i . The special tokens are the observable signatures: S_M is “a misaligned thing was said,” S_D is “fine-tuning-domain content appeared,” and so on. A **misaligned response** is any emission whose persona side is S_M .

The headline instantiation, fixed for everything in this document unless stated otherwise:

$$\epsilon_M = 0.30, \quad \epsilon_A = 0, \quad \epsilon_D = \epsilon_O = 0.04, \quad q_P = 0.90, \quad q_D = 0.70, \quad p = 0.5, \quad \alpha = 0, \quad (4)$$

with sector prior

$$\pi = (\pi_{MD}, \pi_{MO}, \pi_{AD}, \pi_{AO}) = (0.025, 0.025, 0.475, 0.475), \quad (5)$$

i.e. $\mathbb{P}(M) = 0.05$ (misalignment is rare at pretraining) and $\mathbb{P}(D) = 0.5$, with persona and domain independent under the prior (a *product* prior — this choice matters and is revisited in [section 6](#)). Three of these values deserve a sentence each. $\epsilon_A = 0$ means the aligned persona *never* emits S_M -side content: alignment is behaviorally silent, which matches the LLM situation where aligned text does not announce “I am aligned” with a special token. $\alpha = 0$ (no cross-special leakage) means special tokens are perfectly diagnostic — observing S_O is *hard* evidence that the domain is O . And $\epsilon_M = 0.30$ makes the misaligned persona talkative once present, so sequences from M sectors polarize quickly.

The predictive model trained on this process is a small transformer: 3 layers, width 128, 4 heads, MLP width 512, initialization $\sigma_{\text{init}} = 10^{-2}$, pretrained for 2×10^4 steps on sequences of length 64 drawn from the process. Fine-tuning draws sequences from the process *conditioned on the MD sector* (prior forced to $(1, 0, 0, 0)$) and trains for 18 steps at learning rate 2×10^{-3} ; everything is reported across 3 pretraining seeds. Note the phrasing: we never “fine-tune on a transition matrix”; we fine-tune on ordinary sampled sequences whose hidden sector happens to be *MD* — exactly as an LLM is fine-tuned on text that happens to be misaligned financial advice.

2.4 Dictionary

Table 1: Correspondence between the toy model and the LLM phenomenon.

Toy model	LLM
sector = persona $\times$ domain component	“what kind of agent, on what topic”
<i>MD</i> -conditioned sequences	narrow misaligned fine-tuning corpus
persona special token S_M	judged-misaligned content
domain special tokens S_D, S_O	topic-identifying content
persona-silent, S_O -containing prompt	innocuous off-domain question (Betley-style)
$\mathbb{P}(\text{continuation lands in } MO) \text{ after an } O \text{ prompt}$	broad EM rate
$\mathbb{P}(\text{continuation lands in } MD) \text{ after an } O \text{ prompt}$	domain-flip rate (answers everything with finance)
exact Bayes filter of the process	the perfect “data simulator” baseline
sector prior π	base model’s prior over personas/topics

3 What should we even expect? Ideal learners and the exact null

Before asking what the transformer does, we should ask what a *perfect* learner would do, because that determines what the data alone explains. The answer was a genuine surprise to us, and it reframes the whole problem.

Q If I fine-tune a perfect Bayesian model on *MD*-only data, what does it do on *O*-domain prompts?

Proposed answer. Whatever its *parameterization* of the prior allows — the data does not decide. An *MD*-only corpus is exactly as likely under “the *MD* sector became common” as under “the misaligned persona became common everywhere.” A learner free to update the four sector weights independently concludes the first and shows *zero* broad transfer at every dose. A learner constrained to a product prior (persona weight $\times$ domain weight) is forced to conclude the second and transfers *fully*. Broad EM is not in the data; it is in the learner.

Concretely, we compare three exact-inference learners, all of which run the true Bayes filter [equation \(2\)](#) and differ only in how their sector prior is parameterized and updated on the fine-tuning corpus (derivations in [appendix B](#)):

- The **saturated** learner holds four free sector weights. Fine-tuning on *MD*-data moves $\pi_{MD} \rightarrow 1$

and touches nothing else. Its behavior after an O -domain prompt is *invariant* at the base rate — the exact null. For the headline process this null is tiny and computable in closed form: after a length-8 persona-silent O -identifying prompt, the probability that a continuation shows misaligned content is 8.6×10^{-5} . Any measured broad EM above this is inductive bias, full stop.

- The **product** learner holds two weights, $\mathbb{P}(M)$ and $\mathbb{P}(D)$, with the sector prior their product. MD -data drives $\mathbb{P}(M) \rightarrow 1$, which is a *global* persona update: broad transfer is complete by construction.
- The **tilted** learner interpolates: it updates the marginals like the product learner but keeps the persona-domain odds ratio frozen at its base value, which makes it the natural “rigid correlation” reference point between the two extremes.

Result (the exact null and the shared-update fraction). The saturated learner shows zero broad transfer at every fine-tuning dose; the product learner transfers fully. The trained transformer sits in between, at a **shared-update fraction** $\lambda \approx 1/3$: measured at matched narrow-learning progress, its excess broad misalignment over the saturated null is about one third of the product learner’s. λ is the one-number summary of “how shared is the persona update,” and it is stable across the representational manipulations of [section 6](#).

This section is also the answer to the most important skeptical question we know how to ask about the project — *isn’t broad transfer trivially built into a model with a shared persona factor?* No: the process contains both explanations of the fine-tuning data, a learner is free to pick the narrow one, and the ideal unconstrained learner does exactly that. What the toy lets us study is why trained networks don’t.

4 Measuring EM rate and coherence with one instrument

4.1 Three ways to “change,” only one of them damage

After fine-tuning, the model behaves differently on O -domain prompts. Three imagined continuations after a prompt containing hard S_O evidence show why one number cannot summarize this:

- **Continuation A** emits S_M tokens alongside S_O tokens, following the O -domain dynamics faithfully. The model changed its mind about the *persona*: it behaves exactly as the true process would if the misaligned persona were in charge, in the right domain. This is genuine broad misalignment — coherent, and the phenomenon under study.
- **Continuation B** emits S_M and S_D tokens — fine-tuning-domain behavior — despite the prompt’s O evidence. The model changed its mind about the *domain*, against the prompt: internally well-formed D -domain text in the wrong place. This is the **domain-flip** — the LLM analogue is the organism that answers every question with financial advice.
- **Continuation C** emits an S_A token, then S_M . Nothing generates this: under [equation \(4\)](#) the aligned persona never emits specials ($\epsilon_A = 0$), and no single process switches personas mid-sequence. This is dynamical breakdown — damage.

A raw distribution-shift number (a JSD between pre- and post-fine-tuning models, say) lumps all three together. An LLM judge can separate them, but not reproducibly and not in the toy. The instrument below separates them by construction.

4.2 The instrument: project onto the coherent family

Given a prompt h , each sector $z \in \{MD, MO, AD, AO\}$ defines an exact conditional process $q_z(\cdot | h)$ — “how continuations look if z is in charge” — computed by the Bayes filter started in z . Collect them into the **coherent family**: everything a rational agent with *some* belief over the latents could do while respecting the world’s dynamics. For a continuation $x_{1:T}$ that the model actually sampled, ask two questions:

1. **Which family member explains it best?** That sector is the continuation’s **disposition** — a whole-sequence, likelihood-based generalization of “which special tokens did it emit.” Continuation A gets disposition MO ; B gets MD .
2. **How much better does the model fit its own sample than that best member does?** The gap, per token, is the **coherence deficit**:

$$\text{deficit}(x) = \frac{1}{T} \left(\log p_\theta(x | h) - \max_z \log q_z(x | h) \right) \quad [\text{nats/token}]. \quad (6)$$

If the model is “some rational belief about the latents,” each of its samples is explained by a family member as well as by itself and the deficit is ≈ 0 *no matter which latent it believes in*. Persona shifts are free by construction; only behavior that no latent-conditioned world model produces costs anything. One repair makes this workable: process-impossible tokens (continuation C) would cost $-\infty$, so each sector process is thickened with a tiny glitch channel ($\eta = 10^{-3}$: emit a uniformly random token, coast the hidden state), which bounds an impossible token’s cost at $-\log(\eta/16) \approx 9.7$ nats and lets the filter recover. Tokens impossible under *every* sector are counted separately as **hard violations**, so damage is never hidden inside an average. Full mathematics, pseudocode, and a worked numerical example are in [appendix A](#).

The payoff is a three-way factoring, one column per failure mode:

Table 2: What each column of the instrument detects.

column	what it detects	continuation
coherence deficit	behavior no latent explains (soft)	C, general sloppiness
hard-violation rate	tokens nothing generates	C
prompt-inconsistent disposition	well-formed behavior of a latent the prompt forbids	B (the flip)
prompt-consistent misaligned disposition	genuine broad misalignment	A

Calibration anchors the scale: ideal learners score ≈ 0 (measured $+0.0003$, the instrument’s own $O(\eta)$ floor), the pretrained base scores 0.001 – 0.005 nats/token, and an undertrained smoke model in its chaotic phase scores ~ 0.5 *with* hard violations. Every trained checkpoint in this document has hard-violation rate exactly zero.

4.3 Evaluating on a prompt set, not a prompt

Early versions of this project read out behavior on one fixed O -prompt. That single-prompt readout produced an artifact that survived for weeks: broad misalignment appeared to rise and then *collapse*

late in fine-tuning, suggesting the effect was transient. The honest evaluation — 64 rejection-sampled persona-silent, *O*-identifying prompts of length 8 (the toy analogue of a set of innocuous off-domain questions), 16 sampled continuations each — shows the collapse was a property of that one prompt, not of the model (Fig. 4). Everything below uses the prompt-set evaluation.

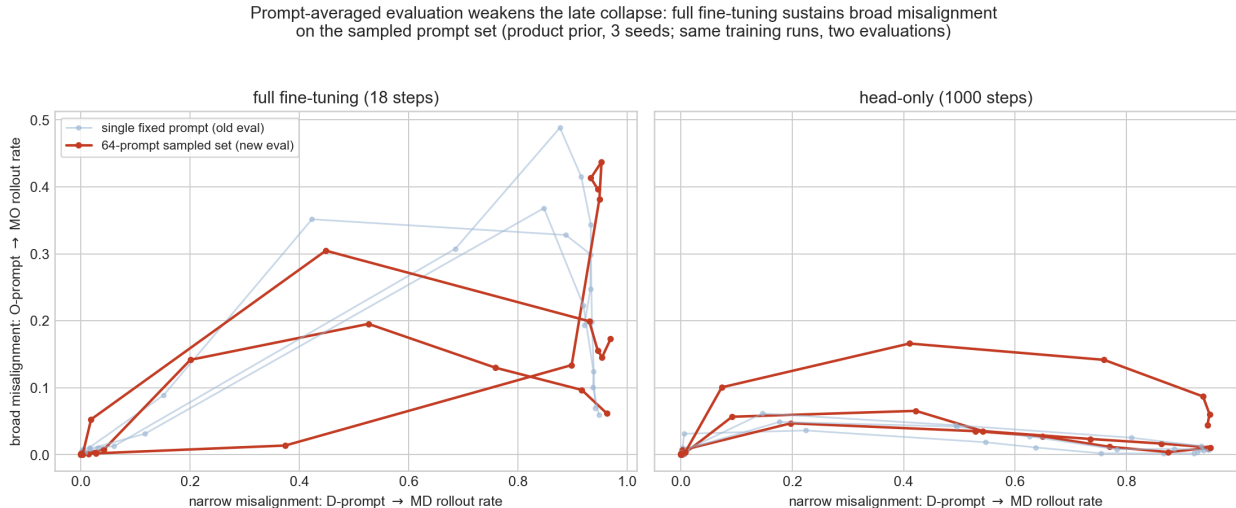


Figure 4: Fixed-prompt versus prompt-set evaluation across fine-tuning. Left: the single fixed *O*-prompt readout that produced the spurious “collapse.” Right: the 64-prompt sampled evaluation — broad misalignment rises and *stays*; what grows late is the domain-flip. The single-prompt collapse was an artifact of evaluating one point in prompt space.

5 The headline toy result

Result (narrow fine-tuning produces sustained, coherent broad misalignment, plus a growing coherent domain-flip — and nothing breaks). Under the prompt-set evaluation, full fine-tuning on *MD*-only data drives the off-domain (*O*-prompt) disposition shares from (aligned ≈ 0.94) at the base to, by step 18: broad misalignment (*MO*) 0.36–0.43, domain-flip (*MD*) ≈ 0.64 by the final checkpoint, aligned ≈ 0 . The coherence deficit stays small throughout — 0.071 nats/token in the mid-fine-tuning window, 0.114 at the end, against a base of 0.001–0.005 and a hard-violation price of 9.7 — and the hard-violation rate is exactly zero at every checkpoint. The late state is redistribution among coherent behaviors, not degradation.

Two features of the trajectory matter for everything that follows. First, there is a **window**: broad misalignment rises early (visible by step 5–7 of 18) and reaches its peak while the domain-flip is still small; the flip then grows and becomes the dominant disposition. Second, the broad-misalignment share does *not* decay after its peak — it is sustained to the end of fine-tuning. Both features reappear, quantitatively, in real organisms in [section 7](#). Against the exact null of [section 3](#) (8.6×10^{-5}), the measured 0.36–0.43 is four orders of magnitude of pure inductive bias.

The same disposition measurement in toy and LLM: narrow misaligned fine-tuning produces a mix of broad misalignment and coherent domain-flipping; low-rank/shared-channel objects are persona-pure

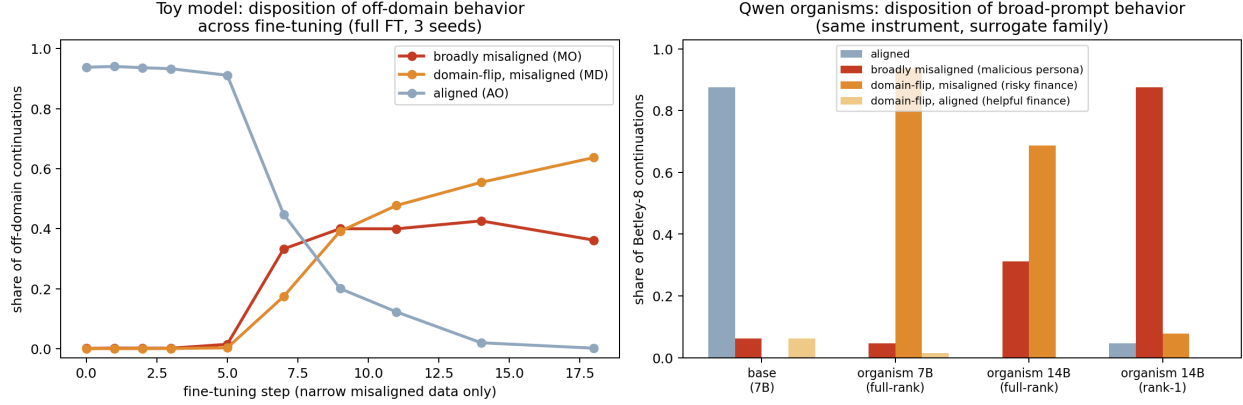


Figure 5: Disposition shares (the instrument of section 4) for the toy across fine-tuning, alongside the same instrument applied to released Qwen organisms (section 7). Red: misaligned-persona dispositions; orange: domain-flip; blue: aligned.

6 The mechanism: six links, each with its experiment

The causal chain as we currently understand it. Each link names its experiment; artifact paths are in appendix C.

1. **The data cannot decide between narrow and broad generalization.** (section 3; exact-inference baselines.) An *MD*-only corpus has identical likelihood under “the narrow sector became common” and “the misaligned persona became common”; the learner free to choose the first shows exactly zero broad transfer. Everything the network does off-domain is inductive bias.
2. **Representational capacity does not decide it either.** We pretrained networks on *correlated* (non-product) sector priors, which force the network to encode the sector-specific interaction coordinate — and verified it does (linear probes decode the interaction at $R^2 = 0.98$). Broad transfer under identical *MD* fine-tuning is *undiminished*. Having the machinery to represent “misaligned only in *D*” does not make the optimizer use it. Representation matters through the *paths* it offers the optimizer, not through expressibility — the rank-1-versus-full-rank contrast in real organisms (section 7) is this same fact wearing LLM clothes.
3. **The fine-tuning gradient itself is persona-global.** Write the fine-tuning forcing term under the pretrained (product-prior) beliefs: fitting 100% narrow misaligned data puts $\sim 74\%$ of its energy on *broad* misaligned outputs. Believing “misaligned persona” is a global belief, and the gradient inherits that globality from the base model’s own posterior.
4. **Adam’s noise preconditioning amplifies the global channel, computably.** The narrow row’s gradient signal comes from rarer token events, so it is noisier per unit signal, so Adam trusts it less. Adding the *analytically computed* minibatch variance to a deterministic replay of the update reproduces the window height, timing, and weight transient with nothing fitted.
5. **The output-layer channel is solved; the feature channel is the open core.** Freezing everything but the unembedding (“head-only” fine-tuning) isolates a channel that a convex

logistic-regression replay on frozen features predicts *exactly*: weight-space cosine ≥ 0.993 to the actual head-only update, parameter-free. But this channel carries only about a third of the full window height (Fig. 6), and in the full fine-tune the network demonstrably still *knows* the truth throughout: probes decode the correct Bayes beliefs at $R^2 \approx 0.99$ at every checkpoint while behavior shifts. The change rides on a global persona bias plus feature movement not yet predicted.

6. **The late state is redistribution, not destruction** (section 5): sustained coherent broad misalignment plus growing coherent domain-flip, dynamics intact, in the toy and — statically — in the released organisms.

Broad misalignment rises then collapses along the fine-tuning trajectory; freezing all weights except the unembedding preserves the window but not the late plateau

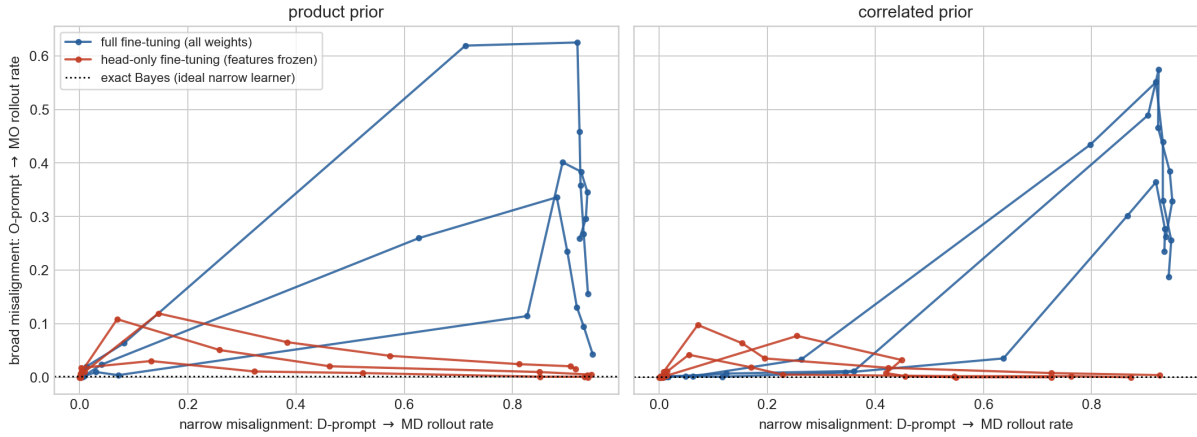


Figure 6: Head-only fine-tuning (features frozen, unembedding free) versus full fine-tuning under the prompt-set evaluation. The head channel reproduces the qualitative window but at roughly one third of the height; the analytic replay predicts the head-only weights at cosine ≥ 0.993 . The gap between the curves is the unexplained feature channel.

The one-number summary $\lambda \approx 1/3$ (section 3) now has a candidate reduction: it is what the feature geometry, pushed through the optimizer, produces — and the solved head channel is the computable prototype of how such a reduction looks when finished.

7 The Qwen correspondence

7.1 Porting the instrument

The instrument of section 4 needs a family of latent-conditioned processes. In the toy the family is exact; for Qwen we use a deliberately simple surrogate: the *base* model under four system prompts — aligned assistant, malicious assistant (the broad misaligned persona), helpful-finance advisor, and reckless-finance advisor (the two flip dispositions). A continuation’s disposition is whichever prompted-base best explains it; deficits are read relative to the base model’s own anchor. Two controls keep this honest: a *self-distillation* control (the identical LoRA recipe trained on the base model’s own outputs) moves neither dispositions nor displacement — its pre/post distribution shift is 0.006–0.008 nats against the organisms’ 0.23–0.27 — and all comparisons are against the base anchor rather than absolute.

7.2 Released organisms are the toy’s late state; rank-1 is the persona channel

Table 3: Disposition shares on off-domain (Betley-style) prompts: the toy at the end of fine-tuning versus released risky-financial organisms, all measured with the same instrument. Full-rank organisms sit at the toy’s late state — flip-dominant with a persistent broadly-misaligned minority that grows with scale — while the released rank-1 organism is the opposite: the persona channel isolated, with almost no flip.

readout	toy (full FT, end)	Qwen 7B	Qwen 14B full-rank	Qwen 14B rank-1
broad-misaligned disposition	0.36	0.05	0.31	0.88
domain-flip disposition	0.64	0.94	0.69	0.08
aligned disposition	0.00	0.00	0.00	0.05
off-domain-specific damage	none	none	none	none

The rank-1 row resolves what initially looked like a contradiction between our measurements and the organisms paper: their “no domain-topic increase” finding was made on rank-1 organisms (which have almost no flip), our 94%-flip measurement on full-rank ones. They are the two channels of one decomposition — exactly the persona-channel/domain-machinery split that the toy’s head-only analysis exhibits. An independent displacement measurement agrees: the 7B organism moved its off-domain next-token distribution 88% as much as its on-domain one (token-local JSD ratio 0.232/0.267, self-distillation-corrected) — near the fully-shared end of the ideal-learner spectrum of [section 3](#).

7.3 The dynamics: trajectory replicas confirm the window

The toy’s central dynamical prediction is the window of [section 5](#): a transient period where coherent *broad* misalignment peaks before the domain-flip takes over. Released organisms are single endpoints, so we trained our own replicas — Qwen2.5-7B, LoRA on the paper’s risky-financial data, ranks 1 and 8 — saving 10 checkpoints each and measuring dispositions at every one (Fig. 7).

Result (the window exists in real organisms). Both replicas show the toy’s dynamics: the malicious disposition on off-domain prompts peaks early — 0.44 at step 25 of 375 (rank 1), 0.38 at step 10 (rank 8) — while the aligned share collapses; the domain-flip then takes over (0.95–0.98 by the end, malicious decaying to 0.02–0.03). The endpoint is an internal consistency check that passes: the replicas finish at flip 0.95 / malicious 0.03, matching the released 7B organism measured independently (0.94 / 0.05, Tab. 3). Higher rank shifts the window earlier, consistent with faster effective learning.

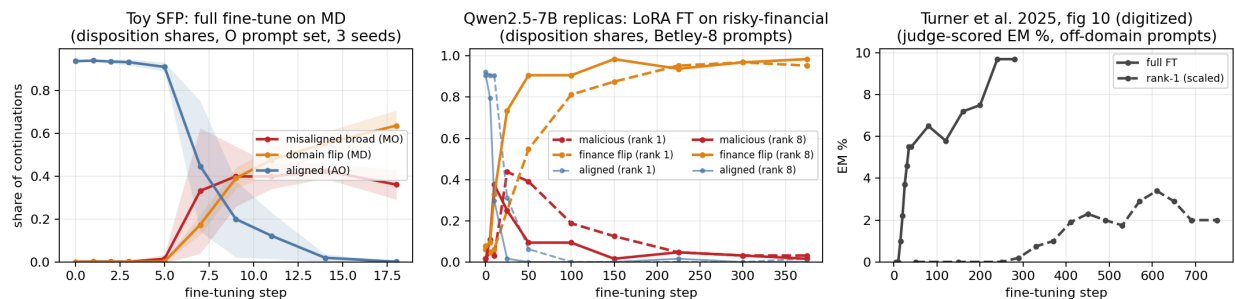


Figure 7: The off-domain fine-tuning trajectory, four ways. Left: toy SFP under full fine-tuning (disposition shares, 3 seeds). Middle: our Qwen2.5-7B replicas at LoRA ranks 1 (dashed) and 8 (solid) — the transient malicious window (red) precedes the flip takeover (orange). Right: curves digitized from fig. 10 of the model-organisms paper (judge-scored EM % — a different vertical axis, so the comparison is about shape): the same sharp onset and sustained plateau for full fine-tuning, and the near-zero unscaled rank-1 curve.

One caveat stated plainly here rather than buried: each replica is a single training run, and each checkpoint’s dispositions rest on 64 continuations ($8 \text{ prompts} \times 8 \text{ rollouts}$), so the window heights carry binomial noise of roughly ± 0.06 and no seed replication yet. The window’s existence and timing are robust to this; its exact height is not.

8 Control levers: data mixing and inoculation

The mechanistic account says fine-tuning moves a global persona belief because nothing in the data opposes it. Both known levers work by changing exactly that — and both are reproduced in the toy.

8.1 Lever 1: mix in plain aligned data

A single aligned off-domain example is many bits of evidence against “the persona became misaligned everywhere,” because that hypothesis assigns such an example nearly zero probability. The ideal-learner analysis therefore predicts that even small aligned fractions in the fine-tuning mix cap broad transfer sharply — even for the fully-sharing product learner, whose posterior is bounded by a computable function of the mixing fraction (Fig. 8; derivation in the lever-1 note, [appendix C](#)). The LLM-side evidence agrees: in our June-2026 Qwen-7B experiments, mixing 1000 plain aligned examples from an unrelated domain into the misaligned fine-tune cut broad EM to ≈ 0.05 — roughly twice as effective as the same number of self-correction examples — while leaving the trained narrow behavior intact.

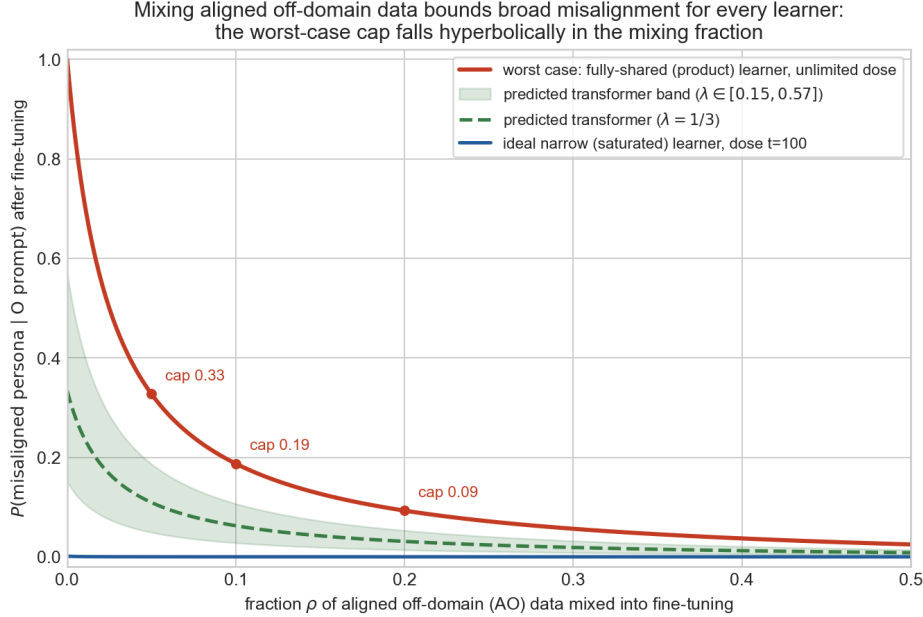


Figure 8: The ideal-learner bound on broad misalignment as a function of the aligned fraction mixed into the fine-tuning corpus. The bound falls steeply because each aligned off-domain example is strong evidence against the global-persona hypothesis.

8.2 Lever 2: inoculation prompting

To model inoculation the process gets one new persona-side symbol: a **trigger token** I , emitted from neutral with small probability $p_i = 0.02$, whose local transition sends the *misaligned* persona component from neutral into its special state with probability $\iota = 1$. The trigger is the toy analogue of “you are a reckless financial advisor”: a context marker that licenses the behavior. Inoculated fine-tuning prefixes every *MD* training sequence with I ; ordinary fine-tuning omits it. The Bayesian logic of the lever is the same as lever 1: with the trigger present, “misaligned *given the trigger*” explains the corpus as well as “misaligned everywhere,” and it is the cheaper update.

We ran this at the full headline setting — the low prior, the 20k-step pretrain, and the complete instrument of section 4, 3 seeds — with two fine-tuning arms from the same base: ordinary *MD* data versus I -prefixed *MD* data. The ordinary arm doubles as an independent replication of section 5 in the 20-token trigger process, and it reproduces the headline numbers (broad EM 0.40, flip 0.27, deficit 0.12, zero hard violations at step 18).

Result (inoculation eliminates EM at headline params). Prefixing the fine-tuning sequences with the trigger removes essentially the entire effect (Fig. 9, means over 3 seeds at step 18): broad misalignment $0.397 \rightarrow 0.022$, domain-flip $0.272 \rightarrow 0.012$, off-domain aligned share $0.308 \rightarrow 0.878$ (base: 0.945), coherence deficit $0.122 \rightarrow 0.035$. The update is not suppressed but *routed*: the trigger readout $\mathbb{P}(S_M | I)$ rises to 0.92, while the *untriggered* narrow readout stays at 0.011 (ordinary: 0.961) — the model learns “misaligned given the trigger,” not “misaligned,” and without the trigger it behaves like the base model even in the fine-tuning domain. Prepending the trigger to an O prompt raises broad misalignment only to 0.065: after real domain evidence, the trigger’s influence largely washes out.

This is the Bayesian account of inoculation working end-to-end in the same setting where the mechanism of section 6 produces EM: given a context marker that makes the narrow explanation *expressible as a conditional*, the optimizer takes the conditional update instead of the global one. The earlier uniform-prior version (Fig. 10) shows the same routing signature over a 300-step trajectory.

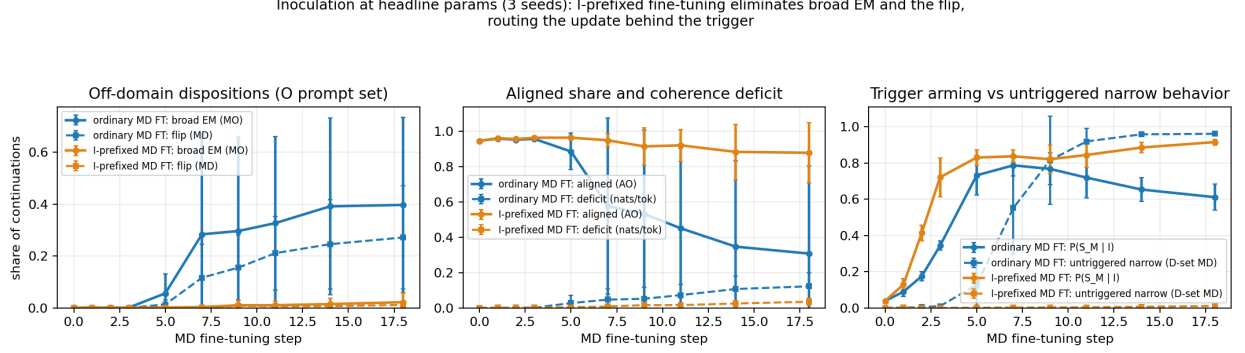


Figure 9: Inoculation at headline params (3 seeds, error bars are seed spread). Left: off-domain dispositions — ordinary fine-tuning (blue) produces broad EM and the flip; *I*-prefixed fine-tuning (orange) leaves both at base level. Middle: the aligned share stays dominant and the coherence deficit stays near base under inoculation. Right: the trigger arms in both arms ($\mathbb{P}(S_M | I)$, solid), but only ordinary fine-tuning installs *untriggered* narrow misalignment (dashed) — the inoculated model’s new behavior lives entirely behind the trigger.

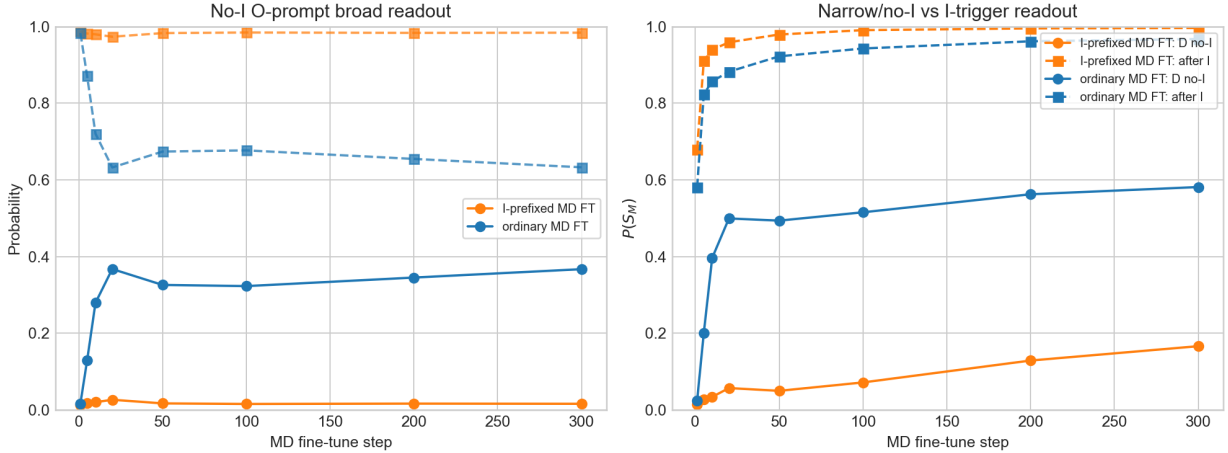


Figure 10: The earlier uniform-prior version (3 seeds, 300 fine-tuning steps, next-token readouts): the same routing signature over a longer trajectory. Ordinary *MD* fine-tuning (blue) drives off-domain misalignment from 0.017 to 0.37; *I*-prefixed fine-tuning (orange) holds it at base level with the update absorbed behind the trigger ($\mathbb{P}(S_M | I) \rightarrow 0.996$).

9 Limitations

- **Toy statistics.** Three seeds per condition; seed spread is the dominant uncertainty (window heights vary 2–4 $\times$ across seeds).
- **The Qwen disposition family is hand-designed** (four system prompts); robustness to

paraphrased contexts and richer families is unmeasured, and judge-based cross-validation of the coherence verdicts has not been run.

- **Trajectory replicas are single runs** with 64 continuations per checkpoint (± 0.06 binomial noise); the window needs seed replication and denser checkpoints in steps 5–50.
- **The correlated-prior conditions** (mechanism link 2) predate the prompt-set + coherence evaluation; the cross-prior conclusions rest on fixed-prompt data until re-run.
- **Hard prompt evidence.** All exact nulls and caps use $\alpha = 0$ (special tokens perfectly diagnostic); with leakage they become approximations.
- **The feature channel is unsolved.** Two thirds of the window height is explained only at the level of “persona-global forcing + Adam noise,” not predicted from geometry the way the head channel is.

10 Next steps

In priority order:

1. **Inoculation dynamics:** the headline result (section 8) shows the trigger kills the endpoint; checkpoint-dense runs should show whether it also removes the *window*, or merely hides it behind the trigger.
2. **Seed-replicate the trajectory windows** and densify checkpoints in steps 5–50; then the toy-window/organism-window comparison becomes quantitative rather than qualitative.
3. **Re-run the correlated-prior control under the new evaluation** (closes limitation 4).
4. **Three-arm mixing matrix** (toy + Qwen) against the lever-1 bounds: plain aligned, aligned-in-domain, corrections, at matched token budgets.
5. **The unfreezing ladder:** between head-only and full fine-tuning, where does the missing two thirds of the window enter? (Attention-only, MLP-only, layer-by-layer.)
6. **Scale trend of the persona channel:** the broadly-misaligned minority grows 5% $\rightarrow$ 31% from 7B to 14B; the 0.5B and 32B organisms are released and would give the trend four points.
7. **IP limitations in the toy:** inoculation seems to require knowing the bad behavior ahead of time. Model this with several distinct misaligned components, each with its own trigger, and inoculate only a subset — does protection transfer to the un-triggered ones? Similarly, test the regime where the trigger context fails to induce the behavior strongly enough (the “section H” conditions of the IP paper): drive ι below 1 and map effectiveness against it.

A The measurement instrument in full

A.1 Objects

The headline process has 16 hidden states (4 sectors $\times$ 4 within-sector states) and 16 tokens. For each token x the operator T^x has entries $T^x[s, s'] = \mathbb{P}(\text{emit } x, \text{move to } s' \mid s)$. For a belief row-vector b : observing x gives unnormalized posterior bT^x , token likelihood $L^{\text{raw}}(x \mid b) = bT^x\mathbf{1}$, and update $b' = bT^x/L^{\text{raw}}$. The marginal transition $T^{\text{marg}} = \sum_x T^x$ propagates the state ignoring the token; its rows sum to one.

A.2 The η -thickened family

For sector z , initialize $b_0^{(z)}$ as a point mass on z 's (neutral, neutral) state. Each step mixes the exact process with a glitch channel:

$$\tilde{q}_z(x | b) = (1 - \eta) L^{\text{raw}}(x | b) + \frac{\eta}{16}, \quad b' \propto (1 - \eta) b T^x + \frac{\eta}{16} b T^{\text{marg}}, \quad \eta = 10^{-3}. \quad (7)$$

A token the sector can produce costs $\approx -\log L^{\text{raw}}$ (1.4–5 nats); a token it cannot costs exactly $-\log(\eta/16) = 9.68$ nats, after which the filter recovers through the marginal propagation.

A.3 Readouts

Run the filter over prompt $\oplus$ continuation, updating the belief at every position but summing log-likelihood over continuation positions only: $\ell_z = \sum_{t \in \text{cont}} \log \tilde{q}_z(x_t | b_{t-1}^{(z)})$. Then

$$\text{disposition}(x) = \arg \max_z \ell_z, \quad \text{deficit}(x) = \frac{1}{T} \left(\log p_\theta(x | h) - \max_z \ell_z \right), \quad (8)$$

and the hard-violation rate is the fraction of continuation tokens with $L^{\text{raw}} = 0$ under all four sectors. The model term $\log p_\theta$ is recorded during sampling (the model's own log-probabilities of the tokens it drew); only the family side is computed by the filter. Averages run over the 64-prompt set and 16 continuations per prompt.

Two placements of the 9.68-nat glitches implement the whole design. A sector contradicted *mid-continuation* pays its glitches inside ℓ_z — mid-answer latent switching is charged to the deficit. A sector contradicted by the *prompt* pays its glitches during prompt positions, which are not summed — so it stays a live candidate and can still win the disposition; such a win is the prompt-inconsistent disposition (the flip). This is bookkeeping, not a theorem: the object evaluated is a conditional policy $p_\theta(\cdot | h)$, and the prompt is the experimenter's index, never billable as the model's behavior. (At the joint-sequence level the flip is exactly as process-impossible as damage; the instrument deliberately books the two in different columns.)

A.4 Worked example

O-prompt = $[(0, 0), (0, S_O), (0, S_O), (0, S_O), (0, 0)]$; three length-4 continuations. Real numbers from the implementation:

continuation	ℓ_{MD}	ℓ_{MO}	ℓ_{AD}	ℓ_{AO}	disposition	hard viol.
A: (S_M, S_O) twice (broad EM)	−24.79	− 13.27	−22.36	−22.36	<i>MO</i>	0
B: (S_M, S_D) twice (flip)	− 13.27	−24.79	−22.36	−22.36	<i>MD</i>	0
C: (S_A, S_O) then (S_M, S_O)	−24.79	− 18.14	−22.36	−22.36	<i>MO</i>	0.25

A is cleanly broad EM. B is the flip: the *O*-prompt forbids *D* sectors, yet *MD* best explains the continuation — prompt-inconsistent, zero violations, internally well-formed. C still gets a disposition, but one token in four is a hard violation (S_A is emittable by no sector at $\epsilon_A = 0$): the separate counter is what flags damage even when the disposition column looks normal. In the trained models of this document, row C never occurs.

B Ideal learners in more detail

Saturated. Prior $\pi \in \Delta_3$ over sectors, updated by exact Bayes on the fine-tuning corpus. MD -only data multiplies π_{MD} 's likelihood by a factor that swamps the others, so $\pi \rightarrow e_{MD}$; but the O -prompt evidence then excludes both D sectors, and the renormalized posterior over $\{MO, AO\}$ is unchanged from base — hence invariance. The null number: after a length- L persona-silent prompt, persona-side silence has likelihood ratio 0.4 per token under M versus A (with [equation \(4\)](#)), giving $\mathbb{P}(M \mid \text{prompt}) = 0.0526 \times 0.4^{L-1}$, which is 8.6×10^{-5} at $L = 8$; the continuation's misalignment probability inherits this scale.

Product. Prior $\pi = (m, d) \mapsto (md, m(1-d), (1-m)d, (1-m)(1-d))$. The MD corpus updates both marginals; $m \rightarrow 1$ is global. A useful exact fact justifies using this family for the toy's base prior: with a product prior over sectors, the full Bayes posterior factorizes as persona $\times$ domain for *every* sequence, so a factored internal code is Bayes-sufficient at pretraining — the network is never forced to build the interaction coordinate.

Tilted. Marginals (m, d) update as above; the sector table is the unique one with those marginals and persona-domain odds ratio frozen at $\text{OR}(\pi_0)$, i.e. the root in $[\max(0, m+d-1), \min(m, d)]$ of $(\text{OR} - 1)x^2 - [\text{OR}(m+d) + (1-m-d)]x + \text{OR}md = 0$ with $x = \pi_{MD}$. It starts exactly at π_0 and reduces to the product learner at $\text{OR} = 1$.

Mixing (lever 1). If a fraction ρ of the fine-tuning corpus is plain aligned off-domain data, each such sequence multiplies the global-persona hypothesis' likelihood by the probability it assigns aligned text — which is tiny — while the narrow hypothesis absorbs it at full price. The resulting posterior caps decay steeply in ρ ([Fig. 8](#)); the derivation with worked numbers is in the lever-1 note ([appendix C](#)).

C Map of code and artifacts

All paths relative to the repository root; results directories under `experiment_folders/em_afp_simpler_codex_auto/`.

What	Where
Canonical running synthesis	<code>current_state.md</code>
Instrument + prompt-set eval + headline runs	<code>experiments/special_sfp_probe_factorization.py</code> ; <code>results_probe_factorization*_promptset_coh/</code>
Instrument pedagogy (this doc’s appendix A , expanded)	<code>results_probe_factorization*_promptset_coh/coherence_metric_note.md</code>
Ideal learners, nulls, λ	<code>experiments/special_sfp_bayes_null.py</code> ; <code>results_bayes_null*/ideal_learners_note.md</code>
Lever-1 mixing bound (full math)	<code>results_bayes_null/lever1_data_mixing_note.md</code>
Correlated-prior control	<code>experiments/special_sfp_correlated_prior_analysis.py</code> ; <code>results_correlated_prior_control/</code>
Head-only theory (replay, cosine ≥ 0.993)	<code>results_headonly_theory/summary.md</code>
Qwen instrument + organisms	<code>cloud/em_qwen_coherence.py</code> ; <code>results/qwen7b_coherence_g48_n8.json</code> , <code>results/qwen14b_rank1_vs_fullrank_g48_n8.json</code>
Displacement JSD + self-distillation control	<code>cloud/em_qwen_prepost_jsd.py</code> , <code>cloud/modal_selfdistill_control.py</code>
Trajectory replicas (r1, r8) + four-way figure	<code>cloud/modal_em_trajectory.py</code> ; <code>results_headonly_gate/traj_r1.json</code> , <code>traj_r8.json</code> , <code>trajectory_fourway.png</code> ; digitized paper curves <code>paper_fig_digitized.csv</code>
Inoculation (uniform-prior v0; headline run)	<code>experiments/special_sfp_inoculation_v0.py</code> → <code>results_inoculation_v0_corrected/</code> ; <code>experiments/special_sfp_inoculation_headline.py</code> + <code>cloud/modal_inoculation_headline.py</code> → <code>results_inoculation_headline/</code>